

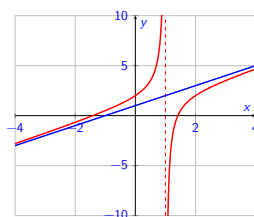
2425-MA140 Engineering Calculus

Week 2, Lecture 2 Introduction to Limits

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This version of the slides are by Niall Madden, but are adapted from original notes by Dr Kirsten Pfeiffer.

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For more, see Section 7.8.1 (Limit of a function of a real number) in *Modern Engineering Mathematics*:

https://search.library.nuigalway.ie/permalink/f/3b1kce/TN_cdi_askewsholts_vlebooks_9780273742517

Tutorials started **this** week. And (I'm really, really, sorry) the **correct** schedule is:

- ▶ Teams 1+2: Tuesday 15:00 ENG-**2003**
- ▶ Teams 3+4: Tuesday 15:00 ENG-**2034** ✓
- ▶ Teams 9+10: Thursday 11:00 ENG-**2002**
- ▶ Teams 11+12: Thursday 11:00 ENG-**3035**
- ▶ Teams 5+6: Friday 13:00 Eng-**2002**
- ▶ Teams 7+8: Friday 13:00 Eng-**2035**

News!

Assignments

- ▶ There is currently a "practice" assignment open. See <https://universityofgalway.instructure.com/courses/35693/assignments/94873>

- ▶ A new assignment will open... **by tomorrow**.

During tutorials, the tutor will solve some similar questions. You can access the tutorial sheet at

https://universityofgalway.instructure.com/courses/35693/files/2023552?module_item_id=650912

In addition, each week I'll post a set of exercises related to the material covered. You don't have to submit your work for these, but you should try them: they are similar in style and standard to exam questions.

Limits

"Not Def" = "not defined".

When we were considering the domain of a function, we looked at those x -values for which the function was not defined.

Example

$$f(x) = \frac{x^2 - 2}{x - 1}$$

$$g(x) = \frac{x^2 - 1}{x - 1}$$

Neither f nor g are defined at $x = 1$.

But what happens if x gets very close to 1?

x	0.900	0.990	0.999	1	1.001	1.010	1.100
$f(x)$	11.9	101.99	1001.99	Not Def	-999.99	-99.99	-9.99
$g(x)$	1.9	1.99	1.999	Not Def	2.001	2.01	2.1

Let's look at the graphs of f and g .

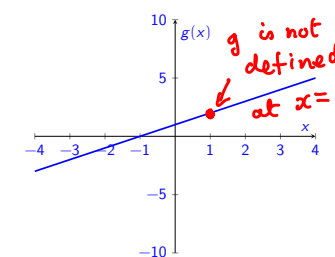
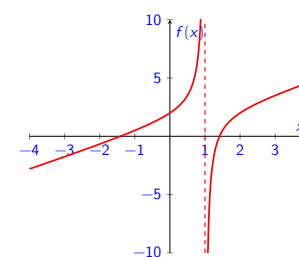
Limits

Except at $x = 1$.

Example

$$f(x) = \frac{x^2 - 2}{x - 1}$$

$$g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x-1)(x+1)}{(x-1)}$$



Limits

In the previous example, we saw that, although neither f nor g was defined at $x = 1$, they behaved very differently as x approaches 1. To discuss this we need some terminology to help us articulate what it means to be really, really close to a value, but not actually at x . We'll also need to be able to discuss what happens for very large or very small x -values.

To do that, we introduce the **limit** L of a function as x approaches some value $a \in \mathbb{R}$ and denote it by

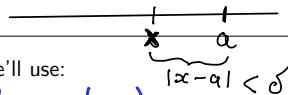
$$\lim_{x \rightarrow a} f(x) = L$$

Note: The concept of a limit is a prerequisite for a proper understanding of calculus and numerical methods.

Definition of a Limit

Some conventions and terminology we'll use:

- ▶ x is a variable. (i.e., a real number)
- ▶ a is a fixed number. (some value of x).
- ▶ ϵ is a small positive number (that we get to choose).
- ▶ δ is another small positive number (determined by ϵ).
- ▶ $|x - a| < \delta$ means that the distance between x and a is less than δ , i.e. very small.
- ▶ As x approaches a , so $f(x)$ approaches a number L .



When we write

$$\lim_{x \rightarrow a} f(x) = L,$$

we read

"The limit of f , as x goes to a , is L ".

Definition of a Limit

LIMIT: formal definition

$$\lim_{x \rightarrow a} f(x) = L,$$

means that, for every number $\epsilon > 0$, it is possible to find a number $\delta > 0$, such that

$$|f(x) - L| < \epsilon \quad \text{whenever} \quad |x - a| < \delta.$$

LIMIT: Informal explanation

$$\lim_{x \rightarrow a} f(x) = L,$$

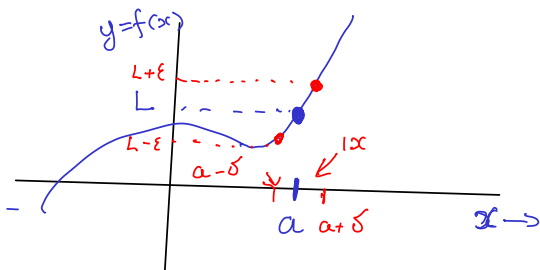
means that we can make $f(x)$ as close to L as we like, by taking x as close to a as needed.

Definition of a Limit

Example

Prove formally that $\lim_{x \rightarrow 3} (4x - 5) = 7$.
That is, for arbitrary ϵ , find a δ such that

$$|(4x - 5) - 7| < \epsilon \quad \text{whenever} \quad |x - 3| < \delta.$$



Definition of a Limit

Example

Prove formally that $\lim_{x \rightarrow 3} (4x - 5) = 7$.
That is, for arbitrary ϵ , find a δ such that

$$|(4x - 5) - 7| < \epsilon \quad \text{whenever} \quad |x - 3| < \delta.$$

$$\begin{aligned} \text{we } & |(4x - 5) - 7| < \epsilon \\ \Rightarrow & |4x - 12| < \epsilon \quad \Rightarrow \quad 4|x - 3| < \epsilon \\ \Rightarrow & |x - 3| < \frac{\epsilon}{4} \\ \text{Since we have } & |x - 3| < \delta, \quad \text{take} \\ & \delta < \frac{\epsilon}{4} \end{aligned}$$

Definition of a Limit

The approach we just used is technically correct, but not very practical in many cases.

Fortunately, there are other methods that can be used

- ▶ to show that a limit exists; ✓
- ▶ find the limit of a function, $f(x)$ as $x \rightarrow a$.

Properties of Limits

Suppose that $\lim_{x \rightarrow a} f_1(x) = L_1$ and $\lim_{x \rightarrow a} f_2(x) = L_2$ and $c \in \mathbb{R}$ is any constant. Then,

(1) $\lim_{x \rightarrow a} c = c, c \in \mathbb{R}$. Eg, if $f(x) = 1$ for all x then $\lim_{x \rightarrow a} f(x) = 1$ for any a .

(2) $\lim_{x \rightarrow a} x = a$. That is, if $f(x) = x$ then $\lim_{x \rightarrow a} f(x) = a$



(3) $\lim_{x \rightarrow a} [cf_1(x)] = cL_1$

Eg $\lim_{x \rightarrow a} [5x] = 5 \lim_{x \rightarrow a} (x) = 5a$.

Properties of Limits

(4) $\lim_{x \rightarrow a} [f_1(x) + f_2(x)] = L_1 + L_2$ and $\lim_{x \rightarrow a} [f_1(x) - f_2(x)] = L_1 - L_2$

Eg $\lim_{x \rightarrow a} (5x + 1) = \lim_{x \rightarrow a} (5x) + \lim_{x \rightarrow a} (1) = 5a + 1$

(5) $\lim_{x \rightarrow a} f_1(x)f_2(x) = L_1L_2$

Eg $\lim_{x \rightarrow a} (x)(x) = \lim_{x \rightarrow a} (x) \cdot \lim_{x \rightarrow a} (x) = (a)(a) = a^2$

(6) $\lim_{x \rightarrow a} (f_1(x))^n = (L_1)^n$

$\lim_{x \rightarrow a} (f_1(x))^n = \underbrace{\left(\lim_{x \rightarrow a} f_1(x) \right) \left(\lim_{x \rightarrow a} f_1(x) \right) \cdots \left(\lim_{x \rightarrow a} f_1(x) \right)}_{n \text{ times}}$

Properties of Limits

(7) $\lim_{x \rightarrow a} \left(\frac{f_1(x)}{f_2(x)} \right) = \frac{L_1}{L_2}$, providing $L_2 \neq 0$.

Eg $\lim_{x \rightarrow 1} \frac{x^2 + 2}{x + 3} = \frac{\lim_{x \rightarrow 1} (x^2 + 2)}{\lim_{x \rightarrow 1} (x + 3)} = \frac{3}{4}$

(8) $\lim_{x \rightarrow a} \sqrt[n]{f_1(x)} = \sqrt[n]{L_1}$

Properties of Limits

Evaluating limits

Note: we can combine these properties as needed. For example, (5) and (8) together give that

$$\lim_{x \rightarrow a} x^n = a^n$$

Example

Evaluate the limit $\lim_{x \rightarrow 1} (x^3 + 4x^2 - 3) = L$

$$\begin{aligned} L &= \lim_{x \rightarrow 1} (x^3) + \lim_{x \rightarrow 1} (4x^2) + \lim_{x \rightarrow 1} (-3) \quad [\text{by (4)}] \\ &= (1)^3 + 4 \underbrace{\lim_{x \rightarrow 1} (x^2)}_{(3)} + \underbrace{(-3)}_{(1)} = 1 + 4 - 3 = 2 \end{aligned}$$

Properties of Limits

Evaluating limits

Example

Evaluate $\lim_{x \rightarrow 1} \frac{x^4 + x^2 - 1}{x^2 + 5} = L$

$$\begin{aligned} \text{Ans } L &= \frac{\lim_{x \rightarrow 1} (x^4 + x^2 - 1)}{\lim_{x \rightarrow 1} (x^2 + 5)} \quad (\text{by (4)}) \\ &= \frac{(1 + 1 - 1)}{1 + 5} = \frac{1}{6} \end{aligned}$$

Limits of rational functions

In many cases it's more complicated. In particular, we'll consider numerous examples where we want to evaluate $\lim_{x \rightarrow a} f(x)$ where a is not in the domain of f .

A typical example of this is when we evaluate a rational function:

$$\lim_{x \rightarrow a} \frac{p(x)}{q(x)}$$

where **both** $p(a) = 0$ and $q(a) = 0$.

Idea: Since we care about the value of p and q near $x = a$, but not actually at $x = a$, it is safe to factor out and $(x - a)$ term from both.

Limits of rational functions

Three examples

Evaluate the limits:

(a) $\lim_{x \rightarrow 0} \frac{x}{x}$

(b) $\lim_{x \rightarrow 0} \frac{x^2}{x}$

(c) $\lim_{x \rightarrow 0} \frac{x}{x^2}$

(a) when $x \neq 0$ $\frac{x}{x} = 1$. So
 $\lim_{x \rightarrow 0} \frac{x}{x} = \lim_{x \rightarrow 0} 1 = 1$ ✓

(b) when $x \neq 0$ $\frac{x^2}{x} = \frac{x}{1} = x$. So $\lim_{x \rightarrow 0} \frac{x^2}{x} = \lim_{x \rightarrow 0} x = 0$.

(c) $\lim_{x \rightarrow 0} \frac{x}{x^2} = \lim_{x \rightarrow 0} \frac{1}{x}$. which is not defined!

Limits of rational functions

Evaluate the limit

$\lim_{x \rightarrow 2} \left(\frac{\frac{1}{2} - \frac{1}{x}}{x - 2} \right)$

Limits of rational functions

Example

Evaluate

$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$

Limits of rational functions

In that last example, we found that

$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x + 2}{x}$

But these are different functions:

More limits

Very often, we'll evaluate limits of the form:

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$

where f and g are not polynomials. Some of the same ideas still apply.

Example

Evaluate

$\lim_{x \rightarrow 0} \frac{\sqrt{1 + x^2} - 1}{x^2}$

Exercises

Exercise 2.2

Evaluate the following limits

(a) $\lim_{x \rightarrow \frac{1}{2}} \frac{x - \frac{1}{2}}{x^2 - \frac{1}{4}}$

(b) $\lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{x^2 + x - 12}$

Exercise 2.3

(From 2023/2024 MA140 exam, Q1(a)) Evaluate the limit

$\lim_{x \rightarrow 4} \frac{x - 4}{(\sqrt{x} - 2)(x + 9)}$